

Rakesh K M

Analog Layout Engineer — Custom IC Layout — AMS/RF Layout — Physical Verification (DRC/LVS) — Cadence Virtuoso

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PROFESSIONAL SUMMARY

- M.Tech VLSI Design graduate with hands-on experience in full-custom Analog Layout using Cadence Virtuoso and Physical Verification using Calibre and Assura DRC/LVS.
- Completed industry-oriented Analog Layout Internship/Training using GlobalFoundries 40nm planar technology, with practical exposure to analog, mixed-signal and CMOS building blocks.
- Worked on full-custom layouts for a Folded-Cascode Low Noise Amplifier (M.Tech Thesis, SCL180nm) and Hybrid Full Adder (GPDK90nm), with hands-on exposure to TSMC 65nm Academic PDK, GlobalFoundries 40nm planar technology, and GPDK45/90/180 through custom analog layout projects and industry-oriented Analog Layout training.
- Skilled in device matching, common-centroid, interdigitation, dummy devices, guard rings, shielding, symmetry, fingering, manual routing, floorplanning, DRC/LVS, PEX and parasitic-aware layout optimization.

TECHNICAL SKILLS

Analog Layout: Full-Custom Layout, Device Matching, Common-Centroid, Interdigitation, Dummy Devices, Guard Rings, Shielding, Symmetry, Fingering, Via Arrays, Manual Routing, Floorplanning, Layout Optimization

Physical Verification: Calibre DRC/LVS, Assura DRC/LVS, RC Parasitic Extraction (PEX), Post-Layout Simulation

Process Technologies / PDKs: GF40nm, TSMC65nm, SCL180nm, GPDK180nm, GPDK90nm, GPDK45nm

EDA Tools: Cadence Virtuoso, Calibre, Assura, Synopsys ICC2, Design Compiler, Synopsys VCS

Analog / Mixed-Signal: Differential Pair, Current Mirror, Single-Stage Op-Amp, Cascode Op-Amp, Comparator, Bandgap Reference, Level Shifter, DAC, Filters, Ring Oscillator, LDO Fundamentals

Analog / RF Analysis: S-parameters, Noise Figure (NF), IIP3, P1dB, K-factor, μ -factor

Programming: C, Java, Python, TCL

ANALOG LAYOUT INTERNSHIP / TRAINING

Semionics — Analog & Mixed-Signal IC Layout & Physical Verification **GF40nm Planar**

- Completed hands-on industry-oriented training in Analog and Mixed-Signal IC Layout Design and Physical Verification using Cadence Virtuoso.
- Implemented differential pairs, current mirrors and passive structures using common-centroid, interdigitation, dummy devices, guard rings, shielding, symmetry and matching-oriented placement.
- Performed DRC/LVS, post-layout parasitic extraction (PEX), routing and layout optimization while following foundry design rules.
- Implemented CMOS blocks including standard cells, AOI gates, clocked NOR-based SR latch, CMOS D flip-flop, 8:3 priority encoder, Modulo-8 counter using JK flip-flops and 4-bit full adder; analog/mixed-signal blocks including single-stage and cascode Op-Amps, multi-voltage level shifter, 4/8-bit DACs, band-pass filter, CMOS bandgap reference, NMOS clamping circuits and ring oscillator.

PROJECTS

Noise-Canceling Low Noise Amplifier (LNA) **2026**

- Designed and analyzed CG-CS, cascode, and folded-cascode LNA topologies in SCL 180nm using Cadence Virtuoso
- Implemented full-custom analog layout using common-centroid placement, interdigitation, dummy devices, symmetry, guard rings and shielding for matching-critical devices.
- Optimized floorplanning, routing and device placement to minimize mismatch and parasitic effects while satisfying analog layout constraints.
- Performed Calibre DRC, LVS, post-layout RC parasitic extraction (PEX), and prepared layouts following foundry design rules and layout verification methodologies.

Low Power Hybrid Full Adder for Array Multiplier **2025**

- Designed hybrid full-adder based 4-bit and 8-bit array multipliers in GPDK90nm CMOS using Cadence Virtuoso.
- Implemented transistor-level schematic and full-custom layout with parasitic-aware layout optimization for low-power operation.
- Performed Assura DRC, LVS and post-layout RC parasitic extraction (PEX).

EDUCATION

Master of Technology (M.Tech) in VLSI Design
Vellore Institute of Technology, Chennai 2024–2026

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering
Sri Sairam Engineering College, Chennai 2019–2023

CERTIFICATIONS

NPTEL

- [System Design Through Verilog](#)
- [VLSI Design Flow: RTL to GDS](#)